Mikkel Paltorp Schmitt

Numerical Acoustics · Scientific Computing · Mathematical Modelling · Programming · Machine Learning

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Summary

I am a self-declared optimist and numbers wizard with a master's degree in mathematical modelling and computation with an emphasis on optimization methods and numerical algorithms. Currently I am enrolled in the PhD program at the Acoustic Technology Group at the Technical University of Denmark. Using my skills in applied mathematics I take Boundary Element simulations to infinity and beyond (or at least close to).

In my spare time I like to go bouldering or sport climbing with friends. When I am not climbing I volunteer as a route setter at my local climbing gym as I enjoy building problems for other climbers to solve. Other stuff that I also enjoy includes bikepacking, various eSports and Formula 1.

Work Experience

Technical University of Denmark *Sep 2020 - Present*

PhD Student - Supervisors: Vicente Cutanda Henriquez & Niels Aage

Efficient Numerical Optimization for Acoustics with Viscothermal Losses and its Application to Acoustic Micro-Devices

- Rewrote the in-house Boundary Element code from MatLab to Julia, reducing the computation time by over two orders of magnitude.
- Collaborations with Prof. Dr.-Ing. Steffen Marburg and PhD Student Simone Preuss at The Technical University of Munich.
- Taught Finite Elements and guided student projects in the course *31265 Numerical acoustics*.

Technical University of Denmark *Sep 2018 - Jan 2020*

Assistant Lecturer (Course 01005)

- Responsibilities included correcting problem sets and supervision/examination of the first year mathematics project.
- Covered topics included complex numbers, linear algebra, (systems of) ODEs, multivariate calculus and simple mathematical modelling.

Education

Technical University of Denmark *Sep 2014 - Jan 2020*

B.Sc.Eng + M.Sc.Eng in Mathematical Modelling and Computation

Specialization within numerical analysis and optimization methods. Took special and PhD- courses in scientific computing and optimization.

I did my thesis on rank-structured matrices and Gaussian processes (Advisor: Martin S. Andersen).

McGill University *Sep 2016 - Dec 2016*

Exchange semester

Took courses in Mathematical Biology, Non-linear Dynamics, Complex Analysis and Honours Partial Differential Equations.

Projects

Special Course with Industry: Methods for Structural Optimization *Collaboration with TICRA*

Together with Martin H. Gæde (Now at TICRA) - Advisors Martin S. Andersen (DTU) & Oscar Borries (TICRA).

- Implemented *the method of moving asymptotes* for minimax optimization in `MatLab` and benchmarked it against TICRAs current solver.

SymSemiseparableMatrices.jl

A package for efficiently computing with symmetric extended generator representable semiseparable matrices

- In this project I learned how to structure a codebase and use version control (`git`). At the same time I gained experience writing automatic tests while using the continuous integration tool `Travis CI`.

Volunteering

Julia Users of Copenhagen *Spring 2022 - Present*

Organizing events with Christian Michelsen

- Our meetups focus on the usage of Julia in industry (PFA Pension, Novozymes, Hafnium Labs) and have had between 25-70 participants.
- The very first meetup included a keynote from the Julia lab Co-PI Chris Rackauckas (in person).

Technical Skills

- Proficient: `Julia` (Including the modelling languages `JuMP` and `Convex.jl`), `MatLab`, `Python`, `macOS`.
- Experience: `COMSOL`, `C`, `C++`, `R`, `CVX`, `MOSEK`, `SQL`, `Git`, `Travis CI`, `Linux`.